Resilient Adaptive Intermittent Control of Nonlinear Systems Under Deception Attacks

Xinjun Wang, Ye Cao, Guangdeng Zong, *Senior Member, IEEE*, Huanqing Wang, and Ben Niu

***Abstract*—In this article, the resilient adaptive intermittent output feedback control is investigated for a class of uncertain strict-feedback nonlinear systems under unknown sensor deception attack. The underlying problem is rather challenging not only because the system contains mismatched uncertainties, the output information is corrupted by an additional attack signal, and only the intermittent false output is available for control design, but also because the virtual control terms are nondifferentiable due to the utilization of event triggering. To circumvent these obstacles, we construct a new fuzzy state estimator via intermittent false output and fuzzy-logic systems (FLSs), which, in conjunction with a new set of error surfaces employing the false and estimated state information, allows a novel observer-based resilient adaptive intermittent control strategy to be developed. The results of the proposed resilient adaptive output event-triggered control shown that the stabilization errors converge to a vicinity of the origin, and all the signals of the overall system remain bounded. Numerical simulations confirm the efficiency of the proposed methodology.**

***Index Terms*—Backstepping, fuzzy-logic systems (FLSs), nonlinear systems, state estimator.**

I. Introduction

THE TOPIC of controller design of complex systems has been a significant discussed in the preceding decades. To handle highly unknown and unmatched nonlinearities, fuzzy-logic systems (FLSs) or neural networks (NNs)-based adaptive control strategy has been frequently paid to the study of design framework for several kinds of nonlinear systems [1], [2], [3], [4], [5], [6], [7], [8]. Recently, with the cross penetration of control, computing, and communication technologies, security of signal transmission has become increasingly important that cannot be ignored and worthy of attention in the field of automation [9], [10]. Under normal circumstances, the attackers can carry out malicious attacks in the following two ways, one is that the attackers find a way to stop the target machine from providing service, i.e., denial-of-service (DOS) attacks [11]. The other is that the attackers try to inject incorrect data, i.e., deception attacks [12], thus the information transmitted by networks is attacked by additive signals. The deception attack is a malicious attack mode in the process of network transmission, which might influence steady-state performance of the systems and leading to instability. Therefore, it is an critical issue to develop resilient control methods to effectively deal with the impact of deception attacks. Until now, some in-depth investigations have been developed for different kinds of systems [12], [13], [14], [15]. The important point is that for the security problems of nonlinear systems under deception attacks, since the attackers destroyed the original state signals, only the wrong states can be received and employed directly to construct controllers. Thus, the issue on resilient adaptive backstepping design for the nonlinear systems with deception attacks is a work which is worth exploring since the traditional error transformation technology mentioned in [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], and [27] will result in the failure to complete the controller design objectives based on backstepping. In addition, when sensors are used for signal transmission in practical applications, considering the limited energy of sensors and the limited bandwidth of transmission channels, it is very necessary and important to decide signal transmission through the event-triggering strategy only when required.

As a more efficient solution for resource utilization, event-triggered strategy, has been received extensive attention in the industrial control process in the past decade. Compared with the conventional time-driven-based control scheme, which is produced continuous control signal to obtain good performance, the event triggering mechanism offers much of the flexibility of achieving the control task by a certain predetermined condition. Therefore, such a new control technique is obviously superior from the perspective of low communication over the time-driven control. To our best knowledge, vast majority of the existing event-triggered control problems for general linear systems are studied in [28], [29], and [30]. However, the majority of physical systems are nonlinear and can be modeled as a set of nonlinear equations, it is more interesting to focus on the event-based control of

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nonlinear system. Fortunately, many representative studies on the event-based control have been extensively discussed for nonlinear systems [31], [32], [33], [34], [35], [36]. By employing adaptive backstepping technique, in [31], a novel event-triggered control method was presented for parametric nonlinear systems in strict-feedback structure under the adaptive event sampling condition. By approximation-based methods, the adaptive control problems were addressed for multi-input multioutput nonlinear systems via event-triggered algorithms in [37]. It is noteworthy that the above works are all assumed that the full states are available. Focusing on the issue of output feedback control under event triggering is undoubtedly meaningful although challenging [38], [39], [40], [41], [42]. To list a few, in [38], when only the output is measurable, the event-based control design was constructed for nonlinear systems with input-to-state stable (ISS) condition. It is important to highlight that the investigated works based on full state feedback or output feedback control methods, the event-triggering architecture is elaborately constructed to refresh control signals only. However, the state/output signals transmitted by sensors are still continuous. To improve the effective utilization of communication channel bandwidth, it is practical to determine the communication between sensor and controller based on event triggering mechanism. Furthermore, considering the cybersecurity problems encountered by the practical applications during signal transmission under deception attacks, the state/output measured by the sensor is corrupted, that is, the false and discontinuous state/output signals are applied to design controllers recursively, rendering the traditional coordinate transformations for Lyapunov-stability-based recursive designs inapplicable.

Motivated by the above limitations, this study proposes the resilient adaptive fuzzy control architecture for a class of strict-feedback nonlinear systems with unknown deception attack in sensor and event-triggered communication. A resilient adaptive fuzzy control scheme is constructed and the technicals difficult in backstepping derivation and stability analysis arising from the nondifferentiable virtual control signals is solved. The contributions are listed as follows.

1) When only output feedback is considered, and the original output of the system becomes unavailable in backstepping design due to the existence of deception attack, it is nontrivial to estimate the unmeasurable states. Here, we propose an effective fuzzy state estimator employing discrete-time false output signal and FLSs, rendering the control design and stability analysis complicated.
2) Different from the previous resilient control results that focus on linear systems with unknown deception attacks [43], [44], here a new coordinate transformation involving error variables with the false output and estimated states is proposed, with which a backstepping design framework utilizing the intermittent false output capable of achieving steady-state performance for the strict-feedback nonlinear systems under deception attacks is proposed.
3) Since the derivative of virtual control needs to be introduced in the backstepping derivation process, the virtual controller designed in each step contains output signal that is not only corrupted by the deception attack, but also discontinuous (nondifferentiable) arisen from triggering. To circumvent this difficulty, by establishing Lemmas 1 and 2, a new design framework of backstepping method using the dynamic filtering technique with continuous false output signal is proposed to compensate for the aforementioned negative impacts effects.

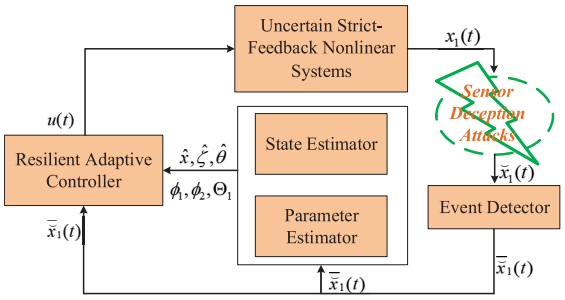


Fig. 1. Control block diagram for nonlinear systems with deception attacks.

II. Problem Formulation and Preliminaries

A. Model Description

In this part, only the triggered output information after attack can be used for controller design. In order to better characterize our proposed controller design and stability analysis method, the following n-order uncertain strict-feedback nonlinear systems is introduced:

$$\begin{aligned}\dot{x}_k &= x_{k+1} + f_k(\bar{x}_k), k = 1, 2, \ldots, n-1\\ \dot{x}_n &= u + f_n(\bar{x}_n)\\ \breve{x}_1 &= x_1 + \omega_s(t, x_1)\\ y &= x_1\end{aligned} \tag{1}$$

where $\bar{x}_i = [x_1, \ldots, x_i]^T \in \mathbb{R}^i$ stand for the system states, $i = 1, 2, \ldots, n$, $y \in \mathbb{R}$ and $u \in \mathbb{R}$ are the system output and input, respectively. $f_i(x_1, \ldots, x_i)$ are the unknown $\mathcal{C}^1$ nonlinear functions. $\omega_s(t, x_1)$ is the sensor deception attack and $\breve{x}_1$ is the available system state after attack.

Fig. 1 shows that the output of the system is attacked by unknown deception. By monitoring the output information of the system, attackers inject malicious deception attack signals into the information transmission path, thus destroying the original output information of the system. In addition, unknown nonlinear functions $f_i(\bar{x}_i)$, $i = 1, 2, \ldots, n$ should be considered in the presence of cyber attacks, which brings additional difficulty in the process of backstepping design, and further discussion will be held later.

In order to save the limited resource utilization in the process of information transmission, the triggering mechanism is proposed as follows:

$$\begin{aligned}\bar{\breve{x}}_1(t) &= \breve{x}_1(t_k), t \in [t_k, t_{k+1})\\ t_{k+1} &= \inf\{t > t_k : |\breve{x}_1(t) - \bar{\breve{x}}_1(t)| \geq m\}\end{aligned} \tag{2}$$

where $t_k > 0$ is the update time and m is a positive constant defined before. During the time $t \in [t_k, t_{k+1})$, the triggered false output information $\bar{\breve{x}}_1(t)$ holds as a constant $\breve{x}_1(t_k)$, once

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the trigger model (2) is triggered, $\bar{\check{x}}_1(t) = \check{x}_1(t_{k+1})$ will be obtained.

The aims of this article are to determine appropriate state estimator and effective resilient adaptive fuzzy control law u for uncertain nonlinear systems (1) by utilizing discrete-time output variable ($\check{x}_1(t_k)$) compromised by attack such that the stabilization errors converge to a small neighborhood of the origin, and all the signals in the closed-loop system are bounded.

To achieve the above-mentioned control purposes, the following assumption is needed.

Assumption 1 [43], [44]: $\omega_s(t, x_1)$ in (1) is described as $\omega_s(t, x_1) = \rho(t)x_1(t)$, where $\rho(t)$ is an unknown time-varying signal, and $1 + \rho(t) > 0$ holds. In addition, $|\rho(t)| \le \bar{\rho}$ and $|\dot{\rho}(t)| \le \bar{\rho}_d$ with $\bar{\rho} > 0$ and $\bar{\rho}_d > 0$ are unknown constants.

Denote $\lambda = (1+\rho)^{-1}$, by recalling Assumption 1, we have $0 < \lambda_m \le \lambda \le \lambda_M$ and $|\dot{\lambda}| \le \bar{\dot{\lambda}}$, where λ_m, λ_M, and $\bar{\dot{\lambda}}$ are unknown positive constants. According to the definition of λ, the formula $x_1 = \lambda \check{x}_1$ holds.

Remark 1: In [45] and [46], the observer-based fault-tolerant control method is developed for a class of nonlinear strict-feedback systems with sensor fault, where the sensor fault ρ is regarded as a constant, and satisfies $0 < \rho \le 1$. However, the deception attack handled in this article is a time-varying signal and additional to system output. In fact, from the mathematical expression form, faults, and attacks are essentially the same, but the malicious attackers will choose the optimal attack strategies, resulting in undetectable attack signals in practice. As a direct result, only the compromised system output $\check{x}_1$ can be used by the designer. Thus, the proposed control strategies in [45] and [46] are no longer applicable, and a new controller design strategy using the compromised error surfaces needs to be considered.

B. Fuzzy-Logic Systems

In this article, FLSs are applied to approximate the unknown functions $f(\eta)$ in the system over a suitable compact set $\Omega_\eta \in R^l$, that is

$$f(\eta) = \phi^{*T}\xi(\eta) + \delta(\eta) \tag{3}$$

where $\phi^* \in R^q$ is the unknown ideal approximating weight matrix and $\eta \in R^l$ is input vector. $\xi(\eta) = [\xi_1(\eta), \ldots, \xi_q(\eta)]^T / \sum_{i=1}^{q} \xi_i(\eta)$ denotes the fuzzy basis function vector with $q > 1$ being the fuzzy rules. δ denotes approximation error.

III. Observer-Based Resilient Adaptive Control

In this section, the designs of observer-based resilient adaptive intermittent control are started for a simple second-order uncertain nonlinear systems (4), and then is extended to the n-order systems

$$\begin{aligned} \dot{x}_1 &= x_2 + f_1(x_1) \\ \dot{x}_2 &= u + f_2(x_1, x_2) \\ \check{x}_1 &= x_1 + \omega_s(t, x_1) \\ y &= x_1 \end{aligned} \tag{4}$$

where $x_1, x_2 \in \mathbb{R}$ stand for the system states, $y \in \mathbb{R}$ and $u \in \mathbb{R}$ are the system output and input, respectively. $f_1(x_1)$ and $f_2(x_1, x_2)$ are the unknown $\mathcal{C}^1$ nonlinear functions. $\omega_s(t, x_1)$ is the sensor deception attack and $\check{x}_1$ is the wrong but available system state.

A. Design of Novel State Estimator

In this article, only the output x_1 can be measured in system (4), therefore, a state observer needs to be developed to realize the estimation of unknown state x_2. However, unlike the general output feedback control method [47], the output x_1 is corrupted owing to the threat of deception attacks. Thus, only the false state $\check{x}_1$ can be used to design the observer. Furthermore, for saving the communication and computation resources, this article considered a more challenge issue that only triggered signal $\bar{\check{x}}_1$ rather than $\check{x}_1$ is available. Under the circumstances, how to design a suitable state estimator is still an open problem.

In addition, note that since $f_i(\bar{x}_i)$, $i = 1, 2$ in (4) are unknown functions, thus the FLSs $\phi_i^T \xi_i(\bar{x}_i)$ is introduced such that $|f_i(\bar{x}_i) - \phi_i^{*T}\xi_i(X_0)| \le \delta_i$, where ϕ_i^* is ideal weight vector, and ϕ_i is the estimation of ϕ_i^*, $X_0 = \bar{x}_i \in \Omega_i$, and Ω_i is a compact set. In this work, we assume that ϕ_i^* and δ_i satisfy $\|\phi_i^*\| \le M_i$ and $|\delta_i| \le \bar{\delta}_i$ with $M_i > 0$ and $\bar{\delta}_i > 0$ being unknown.

To achieve the estimation of unknown state of system (4), a novel fuzzy state estimator is constructed as follows:

$$\begin{aligned} \dot{\hat{x}}_1 &= \hat{x}_2 - l_1\left(\hat{x}_1 - \bar{\check{x}}_1\right) + \hat{f}_1\left(\hat{x}_1|\phi_1\right) \\ \dot{\hat{x}}_2 &= u - l_2\left(\hat{x}_1 - \bar{\check{x}}_1\right) + \hat{f}_2\left(\hat{\bar{x}}_2|\phi_2\right) \end{aligned} \tag{5}$$

where $\hat{x}_i$ is the estimate of x_i, $\hat{f}_i$ is the estimate of f_i, $i = 1, 2$. $L = [l_1, l_2]^T$ with l_1, l_2 is the positive design parameters such that the matrix $A_0 = \left[\begin{smallmatrix} -l_1 & 1 \\ -l_2 & 0 \end{smallmatrix}\right]$ is a Hurwitz matrix. In addition, there exists a positive definite matrix $P > 0$ such that $A_0^T P + P A_0 = -Q$ with $Q > 0$ being a given matrix.

Define $e_i = x_i - \hat{x}_i$, $i = 1, 2$. Then, we have

$$\begin{aligned} \dot{e} &= A_0 e + L\left(x_1 - \hat{x}_1\right) + L\left(\hat{x}_1 - \bar{\check{x}}_1\right) \\ &\quad + \sum_{i=1}^{2} K_i\left(f_i(\bar{x}_i) - \hat{f}_i(\hat{\bar{x}}_i|\phi_i)\right) \\ &= A_0 e + L\left(x_1 - \hat{x}_1\right) + L\left(\hat{x}_1 - \bar{\check{x}}_1\right) + \sum_{i=1}^{2} K_i \omega_i \end{aligned} \tag{6}$$

where $e = [e_1, e_2]^T \in \mathbb{R}^2$, $K_1 = [1, 0]^T$, $K_2 = [0, 1]^T$, and $\omega_i = f_i(\bar{x}_i) - \hat{f}_i(\hat{\bar{x}}_i|\phi_i)$.

Then, the Lyapunov function V_0 is chosen as $V_0 = e^T P e$. According to the triggering mechanism (2), and using (6), Assumption 1, the derivative of V_0 is computed as follows:

$$\begin{aligned} \dot{V}_0 &= -e^T Q e + 2e^T P\left(L(\check{x}_1 - \bar{\check{x}}_1) + \sum_{i=1}^{2} K_i \omega_i - L\omega_s(t, x_{i,1})\right) \\ &\le -e^T Q e + 2\|e\|\|P\|\|L\|m + 2e^T P \sum_{i=1}^{2} K_i \omega_i \\ &\quad + 2\|e\|\|P\|\|L\|\bar{\rho}\lambda_M|\check{x}_1|. \end{aligned} \tag{7}$$

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According to the definitions of ω_i, we have $\omega_i = \phi_i^{*T}(\xi_i(\bar{x}_i) - \xi_i(\hat{\bar{x}}_i)) + \tilde{\phi}_i^T \xi_i(\hat{\bar{x}}_i) + \delta_i$. Then, by using Young's inequality and $0 < \xi_i^T(\cdot)\xi_i(\cdot) \le 1$, the following inequalities hold:

$$e^T P \sum_{i=1}^{2} K_i \omega_i = e^T P \sum_{i=1}^{2} K_i \phi_i^{*T} \left(\xi_i(\bar{x}_i) - \xi_i(\hat{\bar{x}}_i)\right) + e^T P \sum_{i=1}^{2} K_i \tilde{\phi}_i^T \xi_i\left(\hat{\bar{x}}_i\right) + e^T P \sum_{i=1}^{2} K_i \delta_i \le \frac{5\epsilon \|P\|^2}{8} \|e\|^2 + \sum_{i=1}^{2} \left(\frac{\|\tilde{\phi}_i\|^2}{\epsilon} + \frac{4\|\tilde{\phi}_i^*\|^2}{\epsilon} + \frac{2\bar{\delta}_i^2}{\epsilon} \right) \quad (8)$$

$$2\|e\|\|P\|\|L\|m \le \|e\|^2 + \|P\|^2 \|L\|^2 m^2 \quad (9)$$

$$2\|e\|\|P\|\|L\|\bar{\rho}\lambda_M |\check{x}_1| \le \|P\|^2 \|L\|^2 \|e\|^2 + \check{x}_1^2 \bar{\rho}^2 \lambda_M^2. \quad (10)$$

Substituting (8)–(10) into (7) results in

$$\dot{V}_0 \le -e^T Q e + \|P\|^2 \|L\|^2 \|e\|^2 + \|e\|^2 + \frac{5\epsilon \|P\|^2}{4} \|e\|^2 + \frac{2}{\epsilon} \sum_{i=1}^{2} \|\tilde{\phi}_i\|^2 + \check{x}_1^2 \theta + r_0 \quad (11)$$

where $\theta = \bar{\rho}^2 \lambda_M^2$, $r_0 = \|P\|^2 \|L\|^2 m^2 + \sum_{i=1}^{2} (8\|\tilde{\phi}_i^*\|^2/\epsilon + 4\bar{\delta}_i^2/\epsilon)$.

Remark 2: Different from [48], where the system functions $f_i(\cdot)$ only contains the variable x_1, a more general structure, i.e., the strict feedback form is considered in this article. In addition, as stated in Remark 2, owing to the threat of attacks, and only the triggered $\check{\bar{x}}_1$ is introduced for observer design, thus some items $L(\hat{x}_1 - \check{\bar{x}}_1) + \sum_{i=1}^{2} K_i(f_i(\bar{x}_i) - \hat{f}_i(\hat{\bar{x}}_i|\phi_i))$ cannot be processed directly in (6). In order to address these issues, by noting $L(\hat{x}_1 - \check{\bar{x}}_1) = L(\hat{x}_1 - x_1) - L\omega_s(t, x_{i,1}) + L(\check{x}_1 - \check{\bar{x}}_1)$, the middle two items of (6) can be processed as $L(\check{x}_1 - \check{\bar{x}}_1) - L\omega_s(t, x_{i,1})$, and an adaptive attack compensator is constructed to deal with the unknown attack signal $\omega_s(t, x_{i,1})$ later. Another challenge is how to cope with the nonlinear functions $f_i(\bar{x}_i)$, especially in the case of attack, it is difficult to find the relationship between functions before and after triggering, thus the general assumption condition $f_i(y) - f_i(\bar{y}) \le L|y - \bar{y}|$ used in [48] is not suitable. In this article, by applying FLSs approximator to handle the unknown functions $f_i(\bar{x}_i)$, and the adaptive weight updating laws are constructed in the backstepping design process. Meanwhile, considering the triggering mechanism established in (2), the result (11) is finally derived.

B. Resilient Adaptive Fuzzy Control

This part will be focused on solving the issue of resilient adaptive fuzzy control for the system (4) with deception attack. The control design idea is inspired by the dynamic surface design technology, and the following error surfaces are introduced:

$$\begin{aligned} z_1 &= x_1 \\ z_2 &= x_2 - \eta_2 \\ \Xi_2 &= \eta_2 - \alpha_1 \end{aligned} \quad (12)$$

where Ξ_2 is the boundary layer error, α_1 is a virtual control signal, and η_2 is a state variable calculated from the first-order low-pass filters as follows:

$$\kappa_2 \dot{\eta}_2 + \eta_2 = \alpha_1, \eta_2(0) = \alpha_1(0) \quad (13)$$

where $\kappa_2 > 0$.

Remark 3: As shown in Fig. 1, only the output x_1 is measurable. In addition, it should be pointed out that the error surfaces in (12) are only appeared in the process of achieving the stabilization of the closed-loop system based on the Lyapunov method. This is a sharp contrast between the general recursive design framework and the presented recursive design strategy for nonlinear systems subject to sensor deception attack.

Here, a new coordinate transformation of the error is proposed based on the backstepping technique

$$\begin{aligned} \check{z}_1 &= \check{x}_1 \\ \check{z}_2 &= \hat{x}_2 - \eta_2 \end{aligned} \quad (14)$$

where $\check{x}_1$ is an available state signal after attack, which has been defined in (4).

By combining the definition of z_1 and $\check{z}_1$ in (12) and (14), it is easy to get that $z_1 = \lambda \check{z}_1$, and the new error variables $\check{z}_1$ and $\check{z}_2$ will be appeared in the controller. Now, the new backstepping design procedure without considering the triggering mechanism is first presented step by step.

Step 1: The Lyapunov function V_1 is chosen as follows:

$$V_1 = V_0 + \frac{1}{2\lambda} z_1^2 + \frac{1}{2}\Xi_2^2 + \frac{1}{2}\tilde{\zeta}^2 + \frac{1}{2}\tilde{\theta}^2 + \frac{1}{2}\tilde{\phi}_1^T \tilde{\phi}_1 + \frac{1}{2}\tilde{\Theta}_1^2 \quad (15)$$

where $V_0 = e^T P e$, $\tilde{\zeta} = \zeta - \hat{\zeta}$, and $\tilde{\theta} = \theta - \hat{\theta}$ $\tilde{\Theta}_1 = \Theta_1^* - \Theta$. $\zeta = \bar{\lambda}/2$, $\Theta_1^* = \|\phi_1^*\|^2$. $\hat{\zeta}$, $\hat{\theta}$, and Θ are the estimate of ζ, θ, and Θ^*, respectively.

The derivative of V_1 is given as follows:

$$\begin{aligned} \dot{V}_1 &= \dot{V}_0 - \frac{\dot{\lambda}}{2\lambda^2} z_1^2 + \frac{1}{\lambda} z_1 \left(\check{z}_2 + \alpha_1 + \Xi_2 + e_2 + f_1(x_1)\right) + \Xi_2 \dot{\Xi}_2 - \tilde{\zeta}\dot{\hat{\zeta}} - \tilde{\theta}\dot{\hat{\theta}} - \tilde{\phi}_1^T \dot{\phi}_1 - \tilde{\Theta}_1 \dot{\Theta}_1 \\ &= \dot{V}_0 - \frac{\dot{\lambda}}{2\lambda^2} z_1^2 + \frac{1}{\lambda} z_1 (\check{z}_2 + \alpha_1 + \Xi_2 + e_2 + \phi_1^T \xi_1(\hat{x}_1) + \phi_1^{*T} \xi_1(x_1) - \phi_1^{*T} \xi_1(\hat{x}_1) + \tilde{\phi}_1^T \xi_1(\hat{x}_1)) \\ &\quad + \Xi_2 \dot{\Xi}_2 - \tilde{\zeta}\dot{\hat{\zeta}} - \tilde{\theta}\dot{\hat{\theta}} - \tilde{\phi}_1^T \dot{\phi}_1 - \tilde{\Theta}_1 \dot{\Theta}_1. \end{aligned} \quad (16)$$

By using Young's inequality and $0 < \xi_1^T(\cdot)\xi_1(\cdot) \le 1$, we have $\check{z}_1(\phi_1^{*T}\xi_1(x_1) - \phi_1^{*T}\xi_1(\hat{x}_1)) \le (\tau/2)\check{z}_1^2 \Theta_1^* + (2/\tau)$, where $\tau > 0$ is a known constant.

Then, we have

$$\dot{V}_1 \le -e^T Q e + \|P\|^2 \|L\|^2 \|e\|^2 + \|e\|^2 + \frac{5\epsilon \|P\|^2}{4} \|e\|^2 + \frac{2}{\epsilon} \sum_{i=1}^{2} \|\tilde{\phi}_i\|^2 + r_0 + \frac{1}{\lambda} z_1 (\check{z}_2 + \alpha_1 + \Xi_2 + e_2 + \phi_1^T \xi_1(\hat{x}_1) + \frac{\tau}{2} \check{z}_1 \Theta_1^* + \tilde{\phi}_1^T \xi_1(\hat{x}_1)) + \zeta \check{z}_1^2 + \theta \check{z}_1^2 + \Xi_2 \dot{\Xi}_2 - \tilde{\zeta}\dot{\hat{\zeta}} - \tilde{\theta}\dot{\hat{\theta}} - \tilde{\phi}_1^T \dot{\phi}_1 - \tilde{\Theta}_1 \dot{\Theta}_1 + \frac{2}{\tau}. \quad (17)$$

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The virtual controller α_1 is designed as follows:

$$\alpha_1 = -k_1\check{z}_1 - 4\check{z}_1 - \frac{\tau}{2}\check{z}_1 - l_2^2\check{z}_1 - \phi_1^T\xi_1(\hat{x}_1) - \frac{\tau}{2}\check{z}_1\Theta_1 - \hat{\zeta}\check{z}_1 - \hat{\theta}\check{z}_1 \tag{18}$$

where $k_1 > 0$ is a design constant.

By substituting (18) into (17), one can obtain

$$\begin{aligned}\dot{V}_1 \le & -e^TQe + \|P\|^2\|L\|^2\|e\|^2 + \|e\|^2 + \frac{5\epsilon\|P\|^2}{4}\|e\|^2 \\ & + \frac{2}{\epsilon}\sum_{i=1}^{2}\|\tilde{\phi}_i\|^2 + r_0 - \frac{k_1}{\lambda^2}z_1^2 - 2\check{z}_1^2 + \check{z}_1\check{z}_2 + \frac{1}{4}\|e\|^2 \\ & + \frac{1}{2}\Xi_2^2 - \frac{\Xi_2^2}{\kappa_2} - l_2^2\check{z}_1^2 + \tilde{\phi}_1^T\left(\check{z}_1\xi_1(\hat{x}_1) - \dot{\phi}_1\right) \\ & + \tilde{\Theta}_1\left(\frac{\tau}{2}\check{z}_1^2 - \dot{\Theta}_1\right) + \tilde{\zeta}\left(\check{z}_1^2 - \dot{\hat{\zeta}}\right) + \tilde{\theta}\left(\sigma_1 - \dot{\hat{\zeta}}\right) \\ & + \tilde{\theta}\left(\check{z}_1^2 - \sigma_1\right) + \pi_2^2 \end{aligned} \tag{19}$$

where $\pi_2 = -\dot{\alpha}_1$ and σ_1 is a designed variable, which will be given later.

Step 2: Choose the Lyapunov function candidate as $V_2 = (1/2)\check{z}_2^2 + (1/2)\tilde{\phi}_2^T\tilde{\phi}_2$. By using (14), the derivative of V_2 is computed as follows:

$$\begin{aligned}\dot{V}_2 &= \check{z}_2(u + l_2e_1 + l_2(\check{\bar{x}}_1 - \check{x}_1) + l_2\rho(t)\lambda(t)\check{x}_1 + \phi_2^T\xi_2(\hat{\bar{x}}_2) \\ &\quad + \tilde{\phi}_2^T\xi_2(\hat{\bar{x}}_2) - \tilde{\phi}_2^T\xi_2(\hat{\bar{x}}_2) - \dot{\eta}_2) - \tilde{\phi}_2^T\dot{\phi}_2 \\ &\le \check{z}_2(u + l_2e_1 + l_2(\check{\bar{x}}_1 - \check{x}_1) + l_2\rho(t)\lambda(t)\check{x}_1 + \phi_2^T\xi_2(\hat{\bar{x}}_2) \\ &\quad + \tilde{\phi}_2^T\xi_2(\hat{\bar{x}}_2) - \dot{\eta}_2) - \tilde{\phi}_2^T\dot{\phi}_2 + \check{z}_2^2 + \frac{1}{4}\tilde{\phi}_2^T\tilde{\phi}_2. \end{aligned} \tag{20}$$

Until now, if the triggering mechanism is not considered, the general controller is finally designed as $u = \alpha_2, \alpha_2 = -k_2\check{z}_2 - \check{z}_1 - l_2^2\check{z}_2 - \hat{\theta}\check{z}_2 - \phi_2^T\xi_2(\hat{\bar{x}}_2) + \dot{\eta}_2$. In this article, due to the consideration of the trigger mechanism, the compromised state $\check{x}_1$ is not transmitted and only $\check{\bar{x}}_1$ can be obtained and used in constructing the actual controller.

Based on the above analysis, the following error variables are defined

$$\begin{aligned}\check{\bar{z}}_1 &= \check{\bar{x}}_1 \\ \check{\bar{z}}_2 &= \hat{x}_2 - \bar{\eta}_2 \end{aligned} \tag{21}$$

where $\bar{\eta}_2$ can be obtained by adopting the trigger mechanism, i.e., $\bar{\eta}_2(t) = \eta_2(t_{2,k})$, $t \in [t_{2,k}, t_{2,k+1})$, and $t_{2,k+1} = \inf\{t > t_{2,k} : |\eta_2(t) - \bar{\eta}_2(t)| \ge m_{\eta_2}\}$. In this case, the virtual control and the resilient adaptive controller will be redesigned as follows:

$$\bar{\alpha}_1 = -k_1\check{\bar{z}}_1 - 4\check{\bar{z}}_1 - \frac{\tau}{2}\check{\bar{z}}_1 - l_2^2\check{\bar{z}}_1 - \phi_1^T\xi_1(\hat{x}_1) - \frac{\tau}{2}\check{\bar{z}}_1\Theta_1 - \hat{\zeta}\check{\bar{z}}_1 - \hat{\theta}\check{\bar{z}}_1 \tag{22}$$

$$\begin{aligned}u &= \bar{\alpha}_2 \\ \bar{\alpha}_2 &= -k_2\check{\bar{z}}_2 - \check{\bar{z}}_1 - l_2^2\check{\bar{z}}_2 - \hat{\theta}\check{\bar{z}}_2 - \phi_2^T\xi_2\left(\hat{\bar{x}}_2\right) + \dot{\bar{\eta}}_2 \end{aligned} \tag{23}$$

where $\dot{\bar{\eta}}_2 = (\bar{\alpha}_1 - \bar{\eta}_2)/\kappa_2$.

Design the parameter adaptive laws as follows:

$$\begin{aligned}&\dot{\hat{\zeta}} = \check{\bar{z}}_1^2 - \tau_1\hat{\zeta}, \quad \dot{\hat{\theta}} = \sigma_2, \ \sigma_2 = \check{\bar{z}}_2^2 + \sigma_1, \sigma_1 = \check{\bar{z}}_1^2 - \tau_2\hat{\theta} \\ &\dot{\phi}_1 = \check{\bar{z}}_1\xi_1(\hat{x}_1) - s_1\phi_1, \dot{\Theta}_1 = \frac{\tau}{2}\check{\bar{z}}_1^2 - d_1\Theta_1 \\ &\dot{\phi}_2 = \check{\bar{z}}_2\xi_2\left(\hat{\bar{x}}_2\right) - s_2\phi_2 \end{aligned} \tag{24}$$

where τ_1, τ_2, s_1, s_2, and d_1 are positive design parameters.

Remark 4: By comparing α_1 in (18) and $\bar{\alpha}_1$ in (22), it can be seen that $\bar{\alpha}_1$ is designed in a similar form as α_1. The only difference is that the continuous false state signal $\check{x}_1(t)$ involved in α_1 are replaced with piecewise continuous state $\check{\bar{x}}_1(t)$. In such a case, the main difficulty is that the general backstepping framework requires differentiating virtual control inputs recursively. If we chosen the piecewise continuous signals $\bar{\alpha}_1$ in (22) as the virtual controller in step 1, which would lead to the problem that the controller cannot be designed in step 2 due to its derivative does not exist. To solve this problem, the key design idea is to construct normal virtual control α_1 in (18) with continuous signal $\check{x}_1(t)$ in step 1 of recursive design, and then derive continuous control signal α_2, instead of actual control signal $\bar{\alpha}_2$. The effectiveness of this method is verified by proving the boundedness of $\bar{\alpha}_2$ and α_2 in stability analysis.

C. System Stability

In order to demonstrate the stability of the closed-loop system, the Lyapunov function V is constructed as follows:

$$V = V_1 + V_2. \tag{25}$$

From (19) and (20), we have

$$\begin{aligned}\dot{V} \le & -e^TQe + \|P\|^2\|L\|^2\|e\|^2 + \|e\|^2 + \frac{5\epsilon\|P\|^2}{4}\|e\|^2 \\ & + \frac{2}{\epsilon}\sum_{i=1}^{2}\|\tilde{\phi}_i\|^2 + r_0 - \frac{k_1}{\lambda^2}z_1^2 - 2\check{z}_1^2 - k_2\check{z}_2^2 + \frac{1}{4}\|e\|^2 \\ & + \frac{1}{2}\Xi_2^2 - \frac{\Xi_2^2}{\kappa_2} + \tilde{\Theta}_1\left(\frac{\tau}{2}\check{z}_1^2 - \dot{\Theta}_1\right) + \tilde{\zeta}\left(\check{z}_1^2 - \dot{\hat{\zeta}}\right) \\ & + \pi_2^2 + \check{z}_2(-k_2\check{\bar{z}}_2 - \check{\bar{z}}_1 - l_2^2\check{\bar{z}}_2 - \hat{\theta}\check{\bar{z}}_2 + \dot{\bar{\eta}}_2 + k_2\check{z}_2 + \check{z}_1 \\ & + l_2^2\check{z}_2 + \hat{\theta}\check{z}_2 - \dot{\eta}_2 + l_2(\check{\bar{x}}_1 - \check{x}_1)) + \frac{1}{4}\|e\|^2 \\ & + \tilde{\theta}\left(\sigma_2 - \dot{\hat{\theta}}\right) + \tilde{\theta}\left(\check{z}_1^2 + \check{z}_2^2 - \sigma_2\right) + \tilde{\phi}_1^T\left(\check{z}_1\xi_1(\hat{x}_1) - \dot{\phi}_1\right) \\ & + \tilde{\phi}_2^T\left(\check{z}_2\xi_1(\hat{\bar{x}}_2) - \dot{\phi}_2\right) + \check{z}_2^2 + \frac{1}{4}\tilde{\phi}_2^T\tilde{\phi}_2. \end{aligned} \tag{26}$$

To start with the stability analysis, we first propose the following lemma.

Lemma 1: The boundedness between $\check{z}_i$ in (14) and $\check{\bar{z}}_i$ in (21), α_1 in (18) and $\bar{\alpha}_1$ in (22) are summarized as follows:

$$|\check{z}_i - \check{\bar{z}}_i| \le \mu_{\check{z}_i}, i = 1, 2 \tag{27}$$

$$|\alpha_1 - \bar{\alpha}_1| \le \mu_{\alpha_1} + m|\tilde{\zeta}| + m|\tilde{\theta}| + \frac{\tau}{2}m|\tilde{\Theta}| \tag{28}$$

where $\mu_{\check{z}_i}$ and μ_{α_1} are related to event-triggering thresholds m and m_{η_2}, and the design parameter k_1.

Proof: See Appendix A. ∎

Lemma 2: The terms $\tilde{\zeta}(\check{z}_1^2 - \dot{\hat{\zeta}})$, $\tilde{\phi}_1^T(\check{z}_1\xi_1(\hat{x}_1) - \dot{\phi}_1)$, $\tilde{\Theta}_1([\tau/2]\check{z}_1^2 - \dot{\Theta}_1)$, $\tilde{\theta}(\check{z}_1^2 + \check{z}_2^2 - \dot{\hat{\theta}})$, and $\tilde{\phi}_2^T(\check{z}_2\xi_1(\hat{\bar{x}}_2) - \dot{\phi}_2)$ satisfy the following inequalities:

$$\begin{aligned}&\tilde{\zeta}\left(\check{z}_1^2 - \dot{\hat{\zeta}}\right) \le \Delta\zeta, \ \tilde{\phi}_1^T\left(\check{z}_1\xi_1(\hat{x}_1) - \dot{\phi}_1\right) \le \Delta\phi_1 \\ &\tilde{\Theta}_1\left(\frac{\tau}{2}\check{z}_1^2 - \dot{\Theta}_1\right) \le \Delta\Theta_1, \tilde{\theta}\left(\check{z}_1^2 + \check{z}_2^2 - \dot{\hat{\theta}}\right) \le \Delta\theta \\ &\tilde{\phi}_2^T\left(\check{z}_2\xi_1(\hat{\bar{x}}_2) - \dot{\phi}_2\right) \le \Delta\phi_2 \end{aligned} \tag{29}$$

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where $\Delta\zeta, \Delta\phi_1, \Delta\Theta_1, \Delta\theta$, and $\Delta\phi_2$ are related to event-triggering thresholds and the design parameters.

Proof: See Appendix B. ■

Now, the theorem can be obtained as follows.

Theorem 1: Consider nonlinear systems (4) with unknown deception attacks. Under the condition that Assumptions 1 is satisfied, if the state observer (5), the resilient adaptive controller (23), the adaptive update laws (24) are applied, then for any initial conditions satisfying $V(0) \leq \varrho_1$, $\varrho_1 > 0$, all the signals of the closed-loop systems are bounded and the Zeno-behavior does not happen.

Proof: From (26), we have

$$\begin{aligned}
&\check{z}_2(-k_2\bar{\check{z}}_2 - \bar{\check{z}}_1 - l_2^2\bar{\check{z}}_2 - \hat{\theta}\bar{\check{z}}_2 + \bar{\dot{\eta}}_2 + k_2\check{z}_2 + \check{z}_1 + l_2^2\check{z}_2 \\
&\quad + \hat{\theta}\check{z}_2 - \dot{\eta}_2 + l_2(\bar{\check{x}}_1 - \check{x}_1)) \\
&\leq \frac{k_2}{2}\check{z}_2^2 + \frac{k_2}{2}\mu_{\check{z}_2}^2 + \check{z}_2^2 + l_2^4\mu_{\check{z}_2}^2 + \frac{1}{2}\check{z}_2^2 + \frac{1}{2}\mu_{\check{z}_1}^2 + \frac{1}{2}\check{z}_2^2 \\
&\quad + \frac{1}{2}\mu_{\check{z}_2}^2\theta^2 + \frac{1}{2}\check{z}_2^2 + \frac{1}{2}\mu_{\check{z}_2}^2\tilde{\theta}^2 + \frac{1}{2}\check{z}_2^2 + \frac{1}{2\kappa_2^2}\mu_{\alpha_1}^2 + \frac{1}{2}\check{z}_2^2 \\
&\quad + \frac{1}{2\kappa_2^2}m^2\tilde{\zeta}^2 + \frac{1}{2}\check{z}_2^2 + \frac{1}{2\kappa_2^2}m^2\tilde{\theta}^2 + \frac{\tau^2}{8}\check{z}_2^2 + \frac{1}{2\kappa_2^2}m^2\tilde{\Theta}_1^2 \\
&\quad + \frac{1}{2}\check{z}_2^2 + \frac{1}{2\kappa_2^2}\mu_{\check{z}_2}^2 + \frac{1}{2}\check{z}_2^2 + \frac{1}{2}l_2^2m^2. \qquad (30)
\end{aligned}$$

Then, by using Lemmas 1 and 2 and (30), (26) can be rewritten as follows:

$$\begin{aligned}
\dot{V} \leq\ & -e^TQe + \|P\|^2\|L\|^2\|e\|^2 + \|e\|^2 + \frac{5\epsilon\|P\|^2}{4}\|e\|^2 \\
&+ \frac{2}{\epsilon}\sum_{i=1}^{2}\|\tilde{\phi}_i\|^2 + r_0 - \frac{k_1}{\lambda^2}z_1^2 - k_2\check{z}_2^2 + \frac{1}{4}\|e\|^2 + \frac{1}{2}\Xi_2^2 \\
&- \frac{\Xi_2^2}{\kappa_2} + \pi_2^2 + \tilde{\zeta}^2m^2 + \frac{1}{2}\tilde{\zeta}^2 + \frac{1}{2}m^4 - \frac{\tau_1}{2}\tilde{\zeta}^2 + \frac{\tau_1}{2}\zeta^2 \\
&+ \frac{k_2}{2}\check{z}_2^2 + \frac{k_2}{2}\mu_{\check{z}_2}^2 + \frac{c_1}{4}\|\tilde{\phi}_1\|^2m^2 + \frac{1}{c_1} - \frac{s_1}{2}\|\tilde{\phi}_1\|^2 \\
&+ \frac{s_1}{2}\|\phi_1^*\|^2 + \frac{\tau}{2}\tilde{\Theta}_1^2m^2 + \frac{\tau}{4}\tilde{\Theta}_1^2 + \frac{\tau}{4}m^4 - \frac{d_1}{2}\tilde{\Theta}_1^2 \\
&+ \frac{d_1}{2}\Theta_1^{*2} + \frac{c_2}{4}\|\tilde{\phi}_2\|^2m^2 + \frac{1}{c_2} - \frac{s_2}{2}\|\tilde{\phi}_2\|^2 + \frac{s_2}{2}\|\phi_2^*\|^2 \\
&+ \tilde{\theta}^2\mu_{\check{z}_1}^2 + \frac{1}{2}\tilde{\theta}^2 + \frac{1}{2}\mu_{\check{z}_1}^4 + \check{z}_2^2 + \tilde{\theta}^2\mu_{\check{z}_2}^2 + \frac{1}{2}\tilde{\theta}^2 + \frac{1}{2}\mu_{\check{z}_2}^4 \\
&- \frac{\tau_2}{2}\tilde{\theta}^2 + \frac{\tau_2}{2}\theta^2 + \frac{1}{2}\mu_{\check{z}_2}^2\theta^2 + \frac{1}{2}\check{z}_2^2 + \frac{1}{2}\mu_{\check{z}_2}^2\tilde{\theta}^2 + \frac{1}{2}\check{z}_2^2 \\
&+ \check{z}_2^2 + l_2^4\mu_{\check{z}_2}^2 + \frac{1}{2}\check{z}_2^2 + \frac{1}{2}\mu_{\check{z}_1}^2 + \frac{1}{2}\check{z}_2^2 + \frac{1}{2\kappa_2^2}\mu_{\alpha_1}^2 \\
&+ \frac{1}{2}\check{z}_2^2 + \frac{1}{2\kappa_2^2}m^2\tilde{\zeta}^2 + \frac{1}{2}\check{z}_2^2 + \frac{1}{2\kappa_2^2}\mu_{\check{z}_2}^2 + \frac{1}{2}\check{z}_2^2 \\
&+ \frac{1}{2}l_2^2m^2 + \frac{1}{2}\check{z}_2^2 + \frac{1}{2\kappa_2^2}m^2\tilde{\theta}^2 + \frac{\tau^2}{8}\check{z}_2^2 + \frac{1}{2\kappa_2^2}m^2\tilde{\Theta}_1^2 \\
&+ \frac{1}{4}\|e\|^2 + \check{z}_2^2 + \frac{1}{4}\tilde{\phi}_2^T\tilde{\phi}_2. \qquad (31)
\end{aligned}$$

Define the sets $A_1 := \{2e^TPe + z_1^2/\lambda + \check{z}_2^2 + \Xi_2^2 + \tilde{\zeta}^2 + \tilde{\theta}^2 + \tilde{\Theta}_1^2 + \|\tilde{\phi}_1\|^2 + \|\tilde{\phi}_2\|^2 \leq 2\varrho_1\}$ and $B_1 := \{\rho^2 + \dot{\rho}^2 \leq \Lambda_1\}$ with $\Lambda_1 = \bar{\rho}^2 + \bar{\rho}_d^2$. Since $A_1 \times B_1$ is compact in $\mathbb{R}^{10+q_1+q_2}$, there exist a maximum M_2 such that $|\pi_2| \leq M_2$ on $A_1 \times B_1$. Then, we get

$$\begin{aligned}
\dot{V} \leq\ & -\big(\lambda_{\min}(Q) - \|P\|^2\|L\|^2 - \frac{5\epsilon\|P\|^2}{4} - \frac{3}{2}\big)\|e\|^2 + r_0 \\
&- \frac{k_1}{\lambda^2}z_1^2 - \bar{k}_2\check{z}_2^2 - \Gamma\Xi_2^2 + M_2^2 - \bar{\tau}_1\tilde{\zeta}^2 - \bar{\tau}_2\tilde{\theta}^2 - d\tilde{\Theta}_1^2 \\
&- \bar{s}_1\|\tilde{\phi}_1\|^2 - \bar{s}_2\|\tilde{\phi}_2\|^2 + \frac{1}{2}m^4 + \frac{\tau_1}{2}\zeta^2 + \frac{1}{2}\mu_{\check{z}_1}^4 \\
&+ \frac{1}{2}\mu_{\check{z}_2}^4 + \frac{\tau_2}{2}\theta^2 + \frac{1}{2}\mu_{\check{z}_2}^2\theta^2 + \frac{k_2}{2}\mu_{\check{z}_2}^2 + l_2^4\mu_{\check{z}_2}^2 + \frac{1}{2}\mu_{\check{z}_1}^2 \\
&+ \frac{1}{2\kappa_2^2}\mu_{\alpha_1}^2 + \frac{1}{2\kappa_2^2}\mu_{\check{z}_2}^2 + \frac{1}{2}l_2^2m^2 + \frac{1}{c_1} + \frac{s_1}{2}\|\phi_1^*\|^2 \\
&+ \frac{1}{c_2} + \frac{s_2}{2}\|\phi_2^*\|^2 + \frac{\tau}{4}m^4 + \frac{d_1}{2}\Theta_1^{*2} \qquad (32)
\end{aligned}$$

where $\bar{k}_2 = k_2/2 - 7 - \tau^2/8$, $\Gamma = (1/\kappa_2) - (1/2)$, $\bar{\tau}_1 = (\tau_1/2) - m^2 - (1/2) - (1/2\kappa_2^2)m^2$, $\bar{\tau}_2 = (\tau_2/2) - \mu_{\check{z}_1}^2 - \mu_{\check{z}_2}^2 - 1 - (1/2)\mu_{\check{z}_2}^2 - (1/2\kappa_2^2)m^2$, $d = (d_1/2) - (\tau/4) - (1/2\kappa_2^2)m^2$, $\bar{s}_1 = (s_1/2) - (c_1/4)m^2$, and $\bar{s}_2 = (s_2/2) - (c_2/4)m^2 - (1/4)$. Denote $0 < r_1 < \min\{([\lambda_{\min}(Q) - \|P\|^2\|L\|^2 - [(5\epsilon\|P\|^2)/4] - \frac{3}{2}]/\lambda_{\max}(P)), 2k_1/\bar{\lambda}, 2\bar{k}_2, 2\Gamma, 2\bar{\tau}_1, 2\bar{\tau}_2, 2d, 2\bar{s}_1, 2\bar{s}_2\}$, $r_2 = r_0 + M_2^2 + (1/2)m^4 + (\tau_1/2)\zeta^2 + (k_2/2)\mu_{\check{z}_2}^2 + l_2^4\mu_{\check{z}_2}^2 + (1/2)\mu_{\check{z}_1}^2 + (1/2\kappa_2^2)\mu_{\alpha_1}^2 + (1/2\kappa_2^2)\mu_{\check{z}_2}^2 + (1/2)l_2^2m^2 + (1/2)\mu_{\check{z}_1}^4 + (1/2)\mu_{\check{z}_2}^4 + (\tau_2/2)\theta^2 + (1/2)\mu_{\check{z}_2}^2\theta^2 + (1/c_1) + (s_1/2)\|\phi_1^*\|^2 + (1/c_2) + (s_2/2)\|\phi_2^*\|^2 + (\tau/4)m^4 + (d_1/2)\Theta_1^{*2}$. Then, we have

$$\dot{V} \leq -r_1V + r_2. \qquad (33)$$

Thus, $\dot{V} < 0$ on $V = \varrho_1$ when $r_1 > r_2/\varrho_1$. It follows from (33) that $V(t) \leq e^{-r_1t}V(0) + (r_2/r_1)(1 - e^{-r_1t})$. In view of $(1/2\bar{\lambda})z_1^2 \leq V(t)$, the stabilization errors z_1 is exponentially bounded to the compact set $\Omega = \{z_1|z_1| \leq (2\bar{\lambda}r_2/r_1)^{(1/2)}\}$ that can be made arbitrarily small by choosing suitable value r_1. Thus, the signals (i.e., $e, z_1, \check{z}_2, \Xi_2, \tilde{\zeta}, \tilde{\theta}, \tilde{\Theta}_1, \tilde{\phi}_1$, and $\tilde{\phi}_2$) of the closed-loop system are bounded and the stabilization error z_1 converge to a small vicinity around the origin. Using the boundedness of z_1(i.e., x_1) and Assumption 1, $\check{z}_1$(i.e., $\check{x}_1$) is bounded, then according to (23), the control signal u is bounded. Thus, we prove that all the signals of the closed-loop system are bounded.

Next, we prove that the resilient adaptive control design can effectively exclude Zeno behavior [49]. To validate this, we can find a $t^0, t^1 > 0$ satisfying $\forall k, m \in z^+$, $\{t_{k+1} - t_k\} \geq t^0$, $\{t_{m+1} - t_m\} \geq t^1$. Define $\varsigma(t) = \check{x}_1(t) - \bar{\check{x}}_1(t)\ \ \forall t \in [t_k, t_{k+1})$, and $\varsigma_{\eta_2}(t) = \eta_2(t) - \bar{\eta}_2(t)\ \ \forall t \in [t_m, t_{m+1})$, we can obtain

$$\begin{aligned}
\frac{d}{dt}|\varsigma| &= \frac{d}{dt}(\varsigma\cdot\varsigma)^{\frac{1}{2}} = \mathrm{sign}(\varsigma)\dot{\varsigma} \\
&= \mathrm{sign}\Big(\check{x}_1(t) - \bar{\check{x}}_1(t)\Big)\dot{\check{x}}_1 \leq |\dot{\check{x}}_1| \qquad (34) \\
\frac{d}{dt}|\varsigma_{\eta_2}(t)| &= \frac{d}{dt}\big(\varsigma_{\eta_2}(t)\cdot\varsigma_{\eta_2}(t)\big)^{\frac{1}{2}} = \mathrm{sign}\big(\varsigma_{\eta_2}(t)\big)\dot{\varsigma}_{\eta_2}(t) \\
&= \mathrm{sign}(\eta_2 - \bar{\eta}_2)\dot{\eta}_2 \leq |\dot{\eta}_2|. \qquad (35)
\end{aligned}$$

In conjunction with (12), (13), and (18), we can analyze that $\dot{\check{x}}_1$ and $\dot{\eta}_2$ are bounded. Consequently, there exist constants $\varpi > 0$ and $\varpi_{\eta_2} > 0$ satisfying $|\dot{\check{x}}_1(t)| \leq \varpi$, $|\dot{\eta}_2(t)| \leq \varpi_{\eta_2}$. Based on $\varsigma(t_k) = 0$, $\lim_{t\to t_{k+1}}\varsigma(t) = m$, and $\varsigma_{\eta_2}(t_m) = 0$, $\lim_{t\to t_{m+1}}\varsigma_{\eta_2}(t) = m_{\eta_2}$, we can obtain the lower bound of

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TABLE I
CONTROL DESIGN FOR STEP n

Triggering mechanism:
$\bar{\check{x}}_1(t) = \check{x}_1(t_k), t \in [t_k, t_{k+1})$
$t_{k+1} = \inf\{t > t_k : \lvert\check{x}_1(t) - \bar{\check{x}}_1(t)\rvert \geq m\}$
$\bar{\eta}_i(t) = \eta_i(t_{i,k}), t \in [t_{i,k}, t_{i,k+1}), i = 2, \ldots, n$
$t_{i,k+1} = \inf\{t > t_{i,k} : \lvert\eta_i(t) - \bar{\eta}_i(t)\rvert \geq m_{\eta_i}\}$ (a1)
State observer ($m = 1, \ldots, n-1$):
$\dot{\hat{x}}_m = \hat{x}_{m+1} - l_m(\hat{x}_1 - \bar{\check{x}}_1) + \hat{f}_m(\hat{\bar{x}}_m \lvert \phi_m)$
$\dot{\hat{x}}_n = u - l_n(\hat{x}_1 - \bar{\check{x}}_1) + \hat{f}_n(\hat{\bar{x}}_n \lvert \phi_n)$ (a2)
Coordinate transformation:
$z_1 = x_1, \check{z}_1 = \check{x}_1, \check{z}_i = \hat{x}_i - \eta_i, \xi_i = \eta_i - \alpha_{i-1}$ (a3)
First-order low-pass filter:
$\kappa_i \dot{\eta}_i + \eta_i = \alpha_{i-1}, \eta_i(0) = \alpha_{i-1}(0), i = 2, \ldots, n$ (a4)
Virtual control Laws ($i = 2, \ldots, n-1$) :
$\alpha_1 = -k_1 \check{z}_1 - 4\check{z}_1 - \sum_{k=2}^{n} l_k^2 \check{z}_1 - \phi_1^T \xi_1(\hat{x}_1)$ $-\frac{\tau}{2}\check{z}_1 \Theta_1 - \hat{\zeta}\check{z}_1 - \hat{\theta}\check{z}_1$
$\alpha_i = -k_i \check{z}_i - \check{z}_{i-1} - 3\check{z}_i - l_i^2 \check{z}_i - \phi_i^T \xi_i(\hat{\bar{x}}_i) - \hat{\theta}\check{z}_i + \dot{\eta}_i$ (a5)
Control Laws:
$u = -k_n \bar{\check{z}}_n - \bar{\check{z}}_{n-1} - l_n^2 \bar{\check{z}}_n - \phi_n^T \xi_i(\hat{\bar{x}}_n) - \hat{\theta}\bar{\check{z}}_n + \bar{\dot{\eta}}_n$ (a6)
with
$\bar{\check{z}}_1 = \bar{\check{x}}_1, \bar{\check{z}}_i = \hat{x}_i - \bar{\eta}_i, i = 2, \ldots, n$
$\bar{\dot{\eta}}_i = (\bar{\alpha}_{i-1} - \bar{\eta}_i)/\kappa_i, i = 2, \ldots, n$
$\bar{\alpha}_1 = -k_1 \bar{\check{z}}_1 - 4\bar{\check{z}}_1 - \sum_{k=2}^{n} l_k^2 \bar{\check{z}}_1 - \phi_1^T \xi_1(\hat{x}_1)$ $-\frac{\tau}{2}\bar{\check{z}}_1 \Theta_1 - \hat{\zeta}\bar{\check{z}}_1$
$\bar{\alpha}_i = -k_i \bar{\check{z}}_i - \bar{\check{z}}_{i-1} - 2\bar{\check{z}}_i - l_i^2 \bar{\check{z}}_i - \phi_i^T \xi_i(\hat{\bar{x}}_i) - \hat{\theta}\bar{\check{z}}_i + \bar{\dot{\eta}}_i$
Adaptive Laws:
$\dot{\hat{\zeta}} = \bar{\check{z}}_1^2 - \tau_1 \hat{\zeta},\ \dot{\hat{\theta}} = \sigma_n,\ \sigma_i = \bar{\check{z}}_i^2 + \sigma_{i-1};\ i = 2, \ldots, n$
$\sigma_1 = \bar{\check{z}}_1^2 - \tau_2 \hat{\theta}$
$\dot{\phi}_1 = \bar{\check{z}}_1 \xi_1(\hat{x}_1) - s_1 \phi_1, \dot{\Theta}_1 = \frac{\tau}{2}\bar{\check{z}}_1^2 - d_1 \Theta_1$
$\dot{\phi}_i = \bar{\check{z}}_i \xi_i(\hat{\bar{x}}_i) - s_i \phi_i,\ i = 2, \ldots, n$ (a7)

interexecution intervals t^0, t^1 satisfy $t^0 \geq m/\varpi > 0$ and $t^1 \geq m_{\eta_2}/\varpi_{\eta_2} > 0$, which denote that the Zeno-behavior can be avoided efficiently. This completes the proof of Theorem 1. ■

IV. RESILIENT ADAPTIVE CONTROL FOR n-ORDER SYSTEMS

Motivated by the above discussions, this section will be considered a more general nth order nonlinear systems (1). By following the ideas in the above section, the final controller and adaptive laws are included at the final step n, as summarized in Table I. Then, by applying the results obtained in Table I, the following theorem can be given.

Theorem 2: Given the nonlinear systems (1) with unknown deception attacks. Under the condition that Assumption 1 is satisfied, if the state observer (a2), the resilient adaptive controller (a6), the adaptive update law (a7) are applied, then for any initial conditions satisfying $V(0) \leq \varrho_2$, $\varrho_2 > 0$, all the signals of the closed-loop systems are bounded and the Zeno-behavior does not happen.

Proof: Choose the Lyapunov function V_n as $V_n = e^T P e + (1/2\lambda) z_1^2 + \sum_{j=2}^{n} (1/2)\check{z}_j^2 + \sum_{j=2}^{n} (1/2)\Xi_j^2 + (1/2)\tilde{\zeta}^2 + (1/2)\tilde{\theta}^2 + \sum_{j=1}^{n} (1/2)\tilde{\phi}_j^T \tilde{\phi}_j + (1/2)\tilde{\Theta}_1^2$.

Define the sets $A_j := \{2e^T P e + z_1^2/\lambda + \sum_{m=2}^{j}(\check{z}_m^2 + \Xi_m^2) + \tilde{\zeta}^2 + \tilde{\theta}^2 \leq 2\varrho_2\}$ and $B_1 := \{\rho^2 + \dot{\rho}^2 \leq \Lambda_1\}$ with $\Lambda_1 = \bar{\rho}^2 + \bar{\rho}_d^2$. Since $A_j \times B_1$ is a compact, there exist a maximum M_{j+1} such that $\lvert\pi_{j+1}\rvert \leq M_{j+1}$ on $A_j \times B_1$. Then, we get

$$\begin{aligned}
\dot{V} \leq & -\left(\lambda_{\min}(Q) - \|P\|^2\|L\|^2 - \frac{5\epsilon\|P\|^2}{4} - 1 - \frac{n}{4}\right)\|e\|^2 + r_0 \\
& - \frac{k_1}{\lambda^2} z_1^2 - \sum_{j=2}^{n-1} k_j \check{z}_j^2 - \bar{k}_n \check{z}_n^2 - \sum_{j=1}^{n-1} \Gamma_{j+1} \Xi_{j+1}^2 + \sum_{j=1}^{n-1} M_{j+1}^2 \\
& - \bar{\tau}_1 \tilde{\zeta}^2 - \bar{\tau}_2 \tilde{\theta}^2 - d\tilde{\Theta}_1^2 - \sum_{i=1}^{n} \bar{s}_i \|\tilde{\phi}_i\|^2 + \frac{1}{2} m^4 + \frac{\tau_1}{2}\zeta^2 \\
& + \sum_{j=1}^{n}\left(\frac{1}{2}\mu_{\check{z}_j}^4 + \frac{1}{c_j} + \frac{s_j}{2}\|\phi_i^*\|^2\right) + \frac{\tau_2}{2}\theta^2 \\
& + \frac{1}{2}\mu_{\check{z}_n}^2 \theta^2 + \frac{k_n}{2}\mu_{\check{z}_n}^2 + l_n^4 \mu_{\check{z}_n}^2 + \frac{1}{2}\mu_{\check{z}_{n-1}}^2 + \frac{1}{2\kappa_n^2}\mu_{\alpha_{n-1}}^2 \\
& + \frac{1}{2\kappa_n^2}\mu_{\check{z}_n}^2 + \frac{1}{2} l_n^2 m^2 + \frac{\tau}{4} m^4 + \frac{d_1}{2}\Theta_1^{*2}
\end{aligned} \tag{36}$$

where $\bar{k}_n = (k_n - 13)/2$, $\Gamma = (1/[\kappa_{j+1}]) - (1/2)$, $\bar{\tau}_1 = (\tau_1/2) - m^2 - (1/2) - (1/2\kappa_n^2)m^2$, $\bar{\tau}_2 = (\tau_2/2) - \sum_{j=1}^{n} \mu_{\check{z}_j}^2 - (n/2) - (1/2)\mu_{\check{z}_n}^2$, $d = (d_1/2) - (\tau/4) - (1/2\kappa_2^2)m^2$, $\bar{s}_1 = (s_1/2) - (c_1/4)m^2$, $\bar{s}_j = (s_j/2) - (c_j/4)m^2 - (1/4)$, and $j = 2, \ldots, n$. Denote $0 < r_3 < \min\{([\lambda_{\min}(Q) - \|P\|^2\|L\|^2 - (5\epsilon\|P\|^2/4) - 1 - (n/4)]/[\lambda_{\max}(P)]),\ 2k_1/\bar{\lambda},\ 2k_2, \ldots, 2k_{n-1},\ 2\bar{k}_n,\ 2\Gamma,\ 2\bar{\tau}_1,\ 2\bar{\tau}_2,\ 2d,\ 2\bar{s}_1, \ldots, 2\bar{s}_n\}$, $r_4 = r_0 + \sum_{j=1}^{n-1} M_{j+1}^2 + (1/2)m^4 + (\tau_1/2)\zeta^2 + (k_n/2)\mu_{\check{z}_n}^2 + l_n^4 \mu_{\check{z}_n}^2 + (1/2)\mu_{\check{z}_{n-1}}^2 + (1/2\kappa_n^2)\mu_{\alpha_{n-1}}^2 + (1/2\kappa_n^2)\mu_{\check{z}_n}^2 + (1/2) l_n^2 m^2 + \sum_{j=1}^{n} (1/2)\mu_{\check{z}_j}^4 + (\tau_2/2)\theta^2 + (1/2)\mu_{\check{z}_n}^2 \theta^2 + \sum_{j=1}^{n}([1/c_j] + [s_j/2]\|\phi_j^*\|^2) + [\tau/4]m^4 + [d_1/2]\Theta_1^{*2}$.

Then, we have

$$\dot{V} \leq -r_3 V + r_4. \tag{37}$$

According to the similar procedure, Theorem 2 can be achieved. ■

V. SIMULATION STUDIES

A. Numerical Simulation

Consider the following system model:

$$\begin{aligned}
\dot{x}_1 &= x_2 + f_1(x_1) \\
\dot{x}_2 &= u + f_2(x_1, x_2) \\
\check{x}_1 &= x_1 + \omega_s(t, x_1) \\
y &= x_1
\end{aligned} \tag{38}$$

where $f_1(x_1) = 2\cos x_1$ and $f_2(x_1) = 3x_2 \sin x_1$. In simulation, the initial states are selected as $x_1(0) = 0.1$ and $x_2(0) = 0.2$. The initial values of the adaptive laws are selected as $\hat{\zeta}(0) = 0$, $\hat{\theta}(0) = 0$, $\Theta(0) = 0$, $\phi_1(0) = 0$, and $\phi_2(0) = 0$. The attack signal $\rho(t) = 1 + 0.25\cos(t)$. The control values that are required are given as $k_1 = 2$, $k_2 = 15$, $l_1 = 5$, $l_2 = 2$, $\tau_1 = \tau_2 = 0.5$, $\tau = 5$, $s_1 = 0.1$, $s_2 = 0.1$, $m = m_{\eta_2} = 0.03$, and $\kappa_2 = 0.9$.

Simulation curves are exhibited in Figs. 2 and 3, respectively. Fig. 2 gives the trajectories of output x_1, estimated output $\hat{x}_1$, and piecewise continuous state $\check{x}_1$. Fig. 3 displays the control input of u. Based on these simulation results, it

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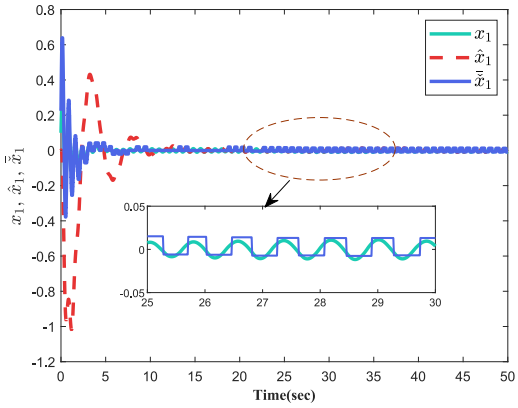


Fig. 2. Trajectories of x_1, $\hat{x}_1$, and $\bar{\hat{x}}_1$.

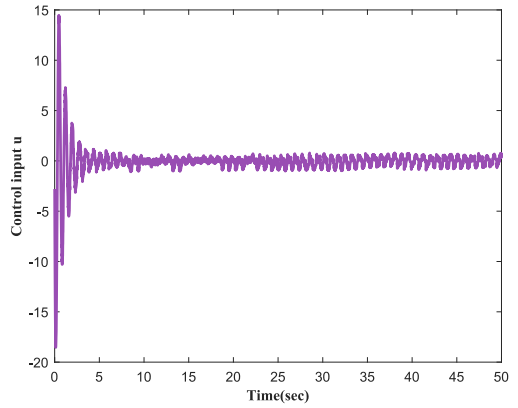


Fig. 3. Response of the control input u.

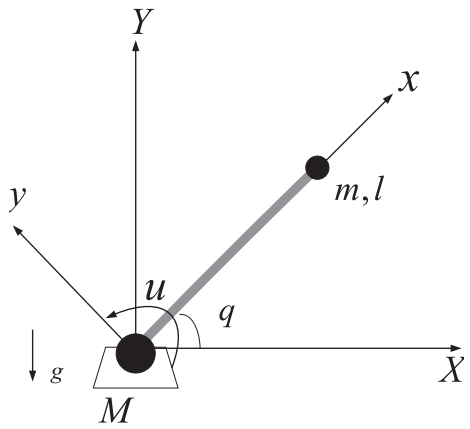


Fig. 4. Framework of single-link robot.

is shown that the proposed resilient security control strategy stabilizes uncertain nonlinear systems with unknown deception attack.

B. Single-Link Robot

The stabilization problem of a single-link robot (see Fig. 4) under sensor attack is simulated in this section, which can be established as follows [50]:

$$M\ddot{q} + \tfrac{1}{2}mgl\sin(q) = u, y = q \tag{39}$$

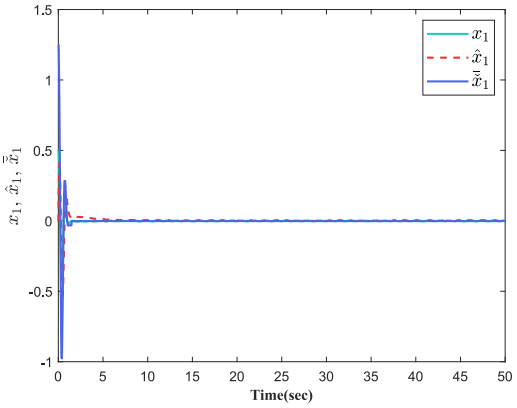


Fig. 5. Trajectories of x_1, $\hat{x}_1$, and $\bar{\hat{x}}_1$.

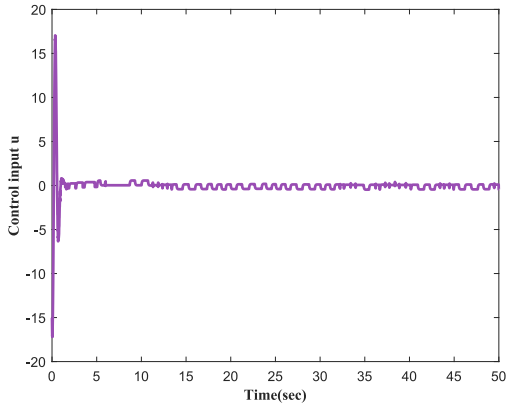


Fig. 6. Response of the control input u.

where $M = 0.5$, $g = 9.8$, $m = 1$, $l = 1$, $q(0) = 0.1$, $\dot{q}(0) = 0.2$, $q_2(0) = 0.3$, $\dot{q}_2(0) = 0.3$, $q_3(0) = 0.4$, and $\dot{q}_3(0) = 0.4$. The physical significance for related parameters can refer to [50]. Let $x_1 = q$, $x_2 = \dot{q}$, then the system model can be rewritten as follows:

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= \left(u - \frac{1}{2}mgl\sin(x_1)\right)/M \\ y &= x_1. \end{aligned} \tag{40}$$

In simulation, the initial states are selected as $x_1(0) = 0.1$ and $x_2(0) = 0.2$. The attack signal $\rho(t) = 1 + 0.5\cos(t)$. The control values that are required are given as $k_1 = 5$, $k_2 = 13$, $l_1 = 5$, $l_2 = 4$, $\tau_1 = \tau_2 = 0.5$, $m = m_{\eta_2} = 0.03$, and $\kappa_2 = 0.05$.

The corresponding simulation results are exhibited in Figs. 5 and 6, respectively. In Fig. 5, the output curves of x_1, $\hat{x}_1$, and $\bar{\hat{x}}_1$ are displayed, which demonstrate good steady-state and estimation performance under sensor attack. Fig. 6 give the control input signal u. Based on the aforementioned simulation results, we can clearly obtain that the proposed resilient adaptive control algorithm ensures that the stabilization errors converges to a small bounded range around origin, and all the signals are bounded.

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VI. Conclusion

This study has investigated an adaptive intermittent control problem via triggered false output for a class of uncertain nonlinear systems against deception attacks. The innovation of this article is that only the intermittent false output is transmitted for the feedback based on event-triggered conditions, and a new fuzzy state estimator is constructed to estimate the actual states. The results of the proposed resilient adaptive output event-triggered control shown that the stabilization errors converge to a small bounded range around origin, and all the signals of the overall system remain bounded. In the future, it will be interesting to relax the assumptions for time-varying attacks and to further investigate the feasibility and stability of intermittent adaptive control under deception attacks when $1+\rho(t)\neq 0$.

Appendix A

Proof of Lemma 1: With the aid of the triggering condition (2), we can directly get $|\check{z}_1-\bar{\check{z}}_1|=|\check{x}_1-\bar{\check{x}}_1|\leq m=:\mu_{\check{z}_1}$. Then, based on α_1 in (18) and $\bar{\alpha}_1$ in (22), we have $|\alpha_1-\bar{\alpha}_1|=|-k_1\check{z}_1-4\check{z}_1-(\tau/2)\check{z}_1-l_2^2\check{z}_1-\phi_1^T\xi_1(\hat{x}_1)-(\tau/2)\check{z}_1\Theta_1-\hat{\zeta}\check{z}_1-\hat{\theta}\check{z}_1+k_1\bar{\check{z}}_1+4\bar{\check{z}}_1+(\tau/2)\bar{\check{z}}_1+l_2^2\bar{\check{z}}_1+\phi_1^T\xi_1(\hat{x}_1)+(\tau/2)\bar{\check{z}}_1\Theta_1+\hat{\zeta}\bar{\check{z}}_1+\hat{\theta}\bar{\check{z}}_1|\leq k_1m+4m+(\tau/2)m+l_2^2m+|\tilde{\zeta}|m+|\zeta|m+|\tilde{\theta}|m+|\theta|m+(\tau/2)m|\tilde{\Theta}_1^*|+(\tau/2)m|\tilde{\Theta}_1|=:\mu_{\alpha_1}+m|\tilde{\zeta}|+m|\tilde{\theta}|+(\tau/2)m|\tilde{\Theta}_1|$, where $\mu_{\alpha_1}=k_1m+4m+(\tau/2)m+l_2^2m+|\zeta|m+|\theta|m+(\tau/2)m|\tilde{\Theta}_1^*|$. Following the similar process based on $\check{z}_2$ in (14) and $\bar{\check{z}}_2$ in (21), we can obtain $|\check{z}_2-\bar{\check{z}}_2|=|\hat{x}_2-\eta_2-\hat{x}_2+\bar{\eta}_2|\leq m_{\eta_2}=:\mu_{\check{z}_2}$. The proof is completed. ■

Appendix B

Proof of Lemma 2: By using Lemma 1, (24) and Young's inequality, we have

$$\begin{aligned}\tilde{\zeta}\left(\check{z}_1^2-\dot{\hat{\zeta}}\right)&=\tilde{\zeta}\left(\check{z}_1^2-\bar{\check{z}}_1^2+\tau_1\hat{\zeta}\right)\\&\leq|\tilde{\zeta}||\check{z}_1|m+|\tilde{\zeta}|\big(|\check{z}_1|+m\big)m-\frac{\tau_1}{2}\tilde{\zeta}^2+\frac{\tau_1}{2}\zeta^2\\&\leq\check{z}_1^2+\tilde{\zeta}^2m^2+\frac{1}{2}\tilde{\zeta}^2+\frac{1}{2}m^4-\frac{\tau_1}{2}\tilde{\zeta}^2+\frac{\tau_1}{2}\zeta^2\triangleq\Delta\zeta.\end{aligned}\tag{41}$$

Similarly, one has

$$\begin{aligned}\tilde{\phi}_1^T\big(\check{z}_1\xi_1(\hat{x}_1)-\dot{\phi}_1\big)&=\tilde{\phi}_1^T\left(\check{z}_1\xi_1(\hat{x}_1)-\bar{\check{z}}_1\xi_1(\hat{x}_1)+s_1\phi_1\right)\\&\leq\|\tilde{\phi}_1\|m\|\xi_1\big(\hat{x}_1\big)\|+s_1\tilde{\phi}_1^T\phi_1\\&\leq\frac{c_1}{4}\|\tilde{\phi}_1\|^2m^2+\frac{1}{c_1}-\frac{s_1}{2}\|\tilde{\phi}_1\|^2+\frac{s_1}{2}\|\phi_1^*\|^2\triangleq\Delta\phi_1\end{aligned}\tag{42}$$

$$\begin{aligned}&\tilde{\Theta}_1\left(\frac{\tau}{2}\check{z}_1^2-\dot{\Theta}_1\right)\\&=\tilde{\Theta}_1\left(\frac{\tau}{2}\check{z}_1^2-\frac{\tau}{2}\check{z}_1\bar{\check{z}}_1+\frac{\tau}{2}\check{z}_1\bar{\check{z}}_1-\frac{\tau}{2}\bar{\check{z}}_1^2+d_1\Theta_1\right)\\&\leq|\tilde{\Theta}_1||\check{z}_1|m\frac{\tau}{2}+|\tilde{\Theta}_1|m\frac{\tau}{2}\big(|\check{z}_1|+m\big)-\frac{d_1}{2}\tilde{\Theta}_1^2+\frac{d_1}{2}\Theta_1^{*2}\\&\leq\frac{\tau}{4}\check{z}_1^2+\frac{\tau}{4}\tilde{\Theta}_1^2m^2+\frac{\tau}{4}\check{z}_1^2+\frac{\tau}{4}\tilde{\Theta}_1^2m^2+\frac{\tau}{4}\tilde{\Theta}_1^2+\frac{\tau}{4}m^4\\&\quad-\frac{d_1}{2}\tilde{\Theta}_1^2+\frac{d_1}{2}\Theta_1^{*2}\\&\leq\frac{\tau}{2}\check{z}_1^2+\frac{\tau}{2}\tilde{\Theta}_1^2m^2+\frac{\tau}{4}\tilde{\Theta}_1^2+\frac{\tau}{4}m^4-\frac{d_1}{2}\tilde{\Theta}_1^2+\frac{d_1}{2}\Theta_1^{*2}\\&\triangleq\Delta\Theta\end{aligned}\tag{43}$$

$$\begin{aligned}\tilde{\theta}\left(\check{z}_1^2+\check{z}_2^2-\dot{\hat{\theta}}\right)&\leq|\tilde{\theta}||\check{z}_1|\mu_{\check{z}_1}+|\tilde{\theta}|\big(|\check{z}_1|+\mu_{\check{z}_1}\big)\mu_{\check{z}_1}\\&\quad+|\tilde{\theta}||\check{z}_2|\mu_{\check{z}_2}+|\tilde{\theta}|\big(|\check{z}_2|+\mu_{\check{z}_2}\big)\mu_{\check{z}_2}-\frac{\tau_2}{2}\tilde{\theta}^2+\frac{\tau_2}{2}\theta^2\\&\leq\check{z}_1^2+\tilde{\theta}^2\mu_{\check{z}_1}^2+\frac{1}{2}\tilde{\theta}^2+\frac{1}{2}\mu_{\check{z}_1}^4+\check{z}_2^2+\tilde{\theta}^2\mu_{\check{z}_2}^2+\frac{1}{2}\tilde{\theta}^2\\&\quad+\frac{1}{2}\mu_{\check{z}_2}^4-\frac{\tau_2}{2}\tilde{\theta}^2+\frac{\tau_2}{2}\theta^2\triangleq\Delta\theta\end{aligned}\tag{44}$$

$$\begin{aligned}\tilde{\phi}_2^T\left(\check{z}_2\xi_1(\bar{\hat{x}}_2)-\dot{\phi}_2\right)&=\tilde{\phi}_2^T\left(\check{z}_2\xi_2(\bar{\hat{x}}_2)-\bar{\check{z}}_2\xi_2(\bar{\hat{x}}_2)+s_2\phi_2\right)\\&\leq\|\tilde{\phi}_2\|m\|\xi_2\left(\bar{\hat{x}}_2\right)\|+s_2\tilde{\phi}_2^T\phi_2\\&\leq\frac{c_2}{4}\|\tilde{\phi}_2\|^2m^2+\frac{1}{c_2}-\frac{s_2}{2}\|\tilde{\phi}_2\|^2+\frac{s_2}{2}\|\phi_2^*\|^2\triangleq\Delta\phi_2.\end{aligned}\tag{45}$$

The proof is completed. ■

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